

Yash Gupta

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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience in Python, Golang, SQL, C++, Embedded C, JavaScript, building backend, cloud, and embedded systems. Focused on scalable APIs, reliable data pipelines, testing, and measurable production outcomes aligned to the role. Additional strengths include security-focused API design a...

TECHNICAL SKILLS

Python, Golang, SQL, PostgreSQL, MySQL, SQLAlchemy, SQLAlchemy ORM, C++, Embedded C, JavaScript, Bash, Flask, REST APIs, Microservices, JWT Authentication, AWS IAM, System Design, Data Structures, Algorithms, API Design

PROFESSIONAL EXPERIENCE

Software Engineer Intern | Nippon Audiotronix Pvt. Ltd.

Sept 2021 – Present

Gurugram

- Architected and developed backend services for Autonomous Emergency Braking (AEB) system using Python and C++, processing 100+ sensor events/second...
- Implemented real-time sensor fusion algorithms combining LiDAR, camera, and radar data with probabilistic filtering techniques for robust object de...
- Reverse-engineered and integrated CAN Bus protocols for vehicle telemetry data pipelines, achieving 99.9% message delivery reliability
- Executed ASPICE-compliant Unit Testing using Unity framework for high-concurrency embedded systems, achieving 95%+ code coverage

PROJECTS

Distributed File Deduplication & Metadata Service on AWS

Nov 2024 - Present

Enterprise-grade hash-based file deduplication service with secure REST APIs and scalable metadata management

Python, AWS Lambda, AWS API Gateway, AWS S3, AWS DynamoDB, AWS IAM, REST APIs, Async/Await

- Designed and implemented hash-based deduplication algorithm reducing storage costs by 60% for duplicate-heavy workloads
- Built serverless REST APIs using API Gateway + Lambda with JWT authentication and role-based access control (RBAC)

CloudMon - Real-Time AWS Resource Monitor Dashboard

Aug 2024 - Oct 2024

Scalable cloud resource usage dashboard with automated cost alerts and ML-based optimization recommendations

Python, Flask, AWS EC2, AWS S3, AWS RDS, Boto3, Docker, PostgreSQL, Jinja2

- Built interactive web dashboard using Flask + Jinja2 displaying real-time EC2, RDS, and S3 resource metrics with live updates
- Implemented Boto3 data fetching pipeline with exponential backoff retry logic for AWS API resilience

Real-Time Data Processing Pipeline for Sensor Telemetry

Mar 2024 - Jul 2024

High-throughput data processing pipeline handling vehicle telemetry with anomaly detection and alerting

Python, Apache Kafka, PostgreSQL, Redis, Unit Testing, CAN Bus

- Architected Kafka consumer application processing 10,000+ messages/second from distributed vehicle sensors
- Implemented anomaly detection using statistical methods and real-time alerting for out-of-threshold sensor readings

EDUCATION

B.Tech in Computer Science & Business Systems | Gyan Ganga Institute of Technology & Sciences

July 2021 – May 2025

Jabalpur | GPA: 8.53/10.0

Coursework: Software Engineering, Operating Systems, Database Management Systems, Computer Networks, Data Structures, Algorithms, Cloud Computing

CERTIFICATIONS

- Oracle Cloud AI Professional – Certified in machine learning and AI on Oracle Cloud
- AWS Cloud Architecting – Specialist in designing scalable AWS infrastructure

ACHIEVEMENTS

- Runner-up, Hackbyte Hackathon 2023 – Developed innovative solution achieving 90%+ accuracy, recognized among 500+ participants
- Shortlisted Round 2, Smart India Hackathon 2023 – Advanced to final round as part of top 1% among 100,000+ teams nationwide